

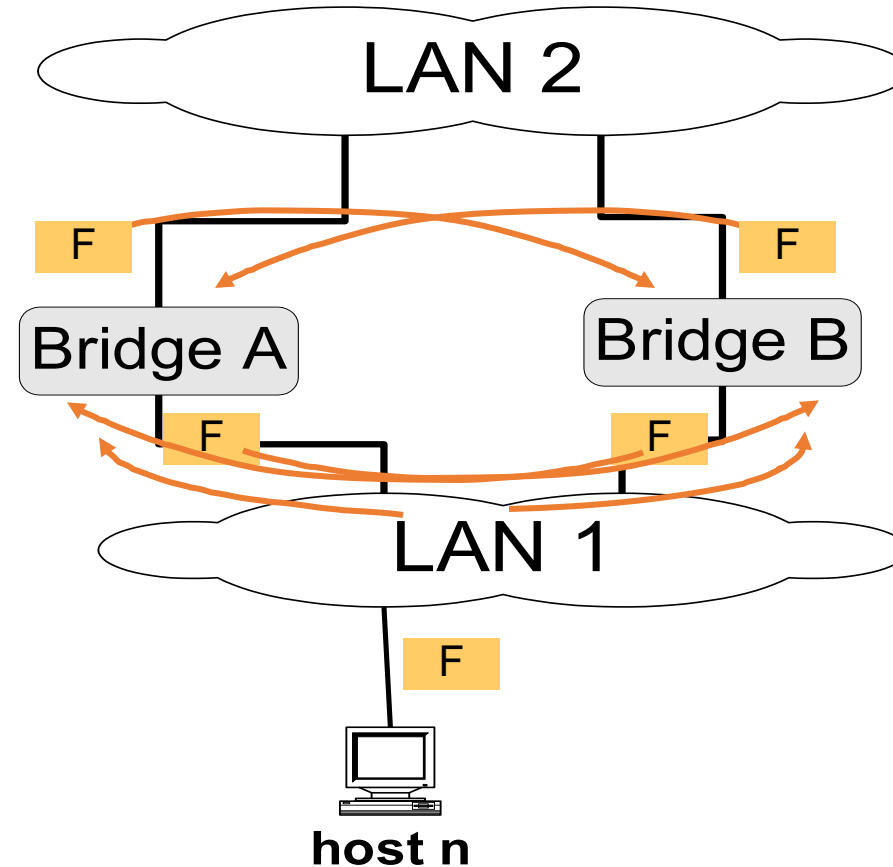
Danger of Loops

- Consider the two LANs that are connected by two bridges.
- Assume *host n* is transmitting a **frame F** with unknown destination.

What happens?

- Bridges A and B flood the frame to LAN 2 and enter LAN 1 as source for host n.
- Bridge B sees F on LAN 2, and copies the frame back to LAN 1 as it has no destination for F.
- Bridge A does the same.
- The copying continues

Where's the problem? What's the solution? **Prevent looping**



Spanning Tree

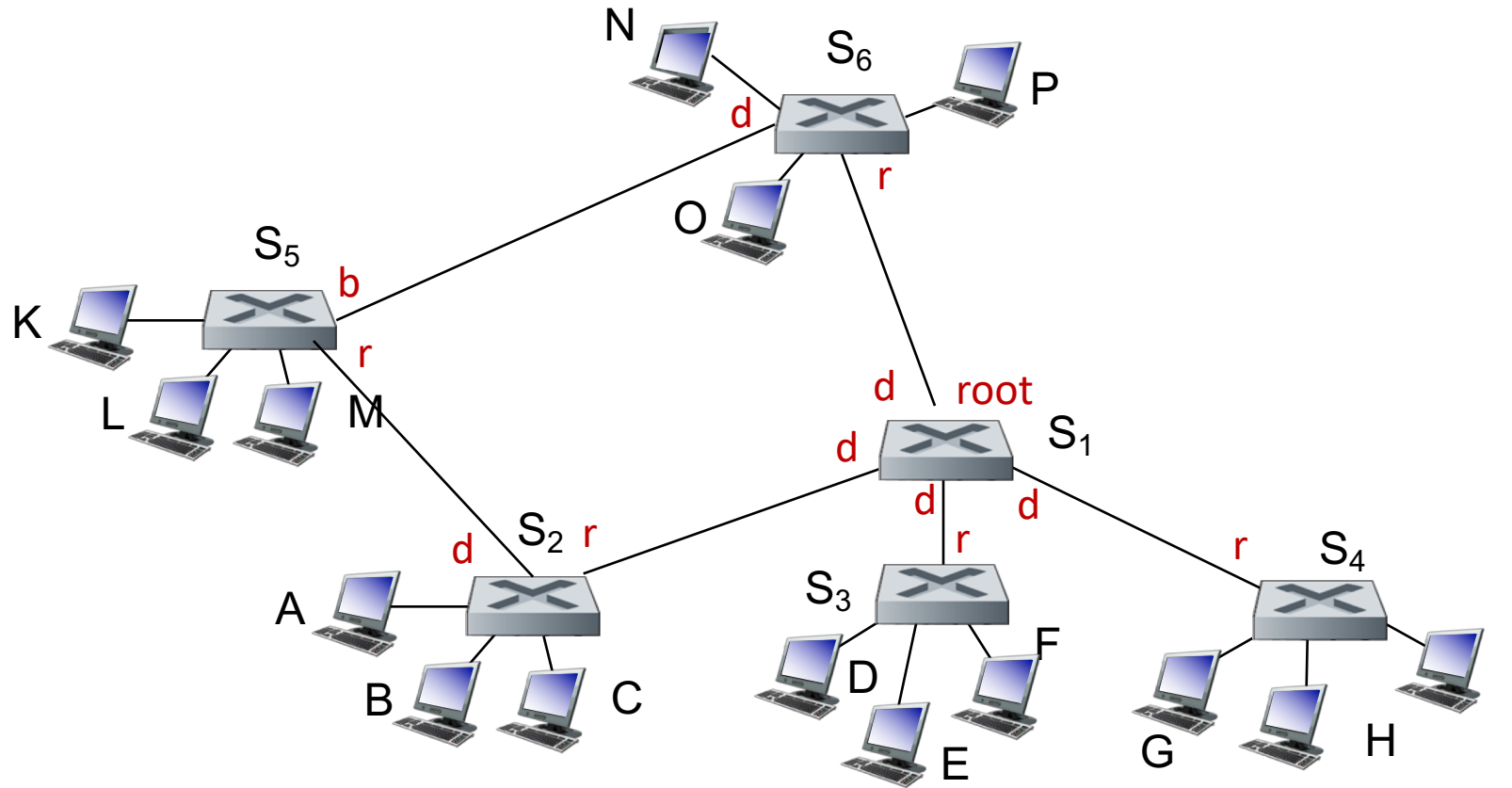
- a solution is to prevent loops in the topology
- IEEE 802.1d standardized an algorithm that builds and maintains a **spanning tree** in a dynamic environment
- switches exchange messages to configure the spanning tree
(**Configuration Bridge Protocol Data Unit** - Configuration BPDU)
- uses flooding, takes sometime to converge
- one bridge elected the root, switches choose shortest path to root
- bridges that run 802.1d are called transparent bridges

BPDUs and Spanning Tree

with the help of BPDUs, switches:

- elect a single switch as the **root switch** (switch with lowest ID)
- calculate the distance of the **shortest path** to the root switch
- a shared medium (broadcast LAN) with multiple switches, picks a **designated switch**, which is the switch with shortest path to root
- ties in shortest paths are broken by choosing a path leading to a switch with lower ID
- each switch has a **root port**, it is the port that leads to the root via the shortest path
- a designated switch will have a **designated port** that will be used for frame forwarding (part of spanning tree)
- switches will forward packets on root and designated ports only
- some ports on a switch will be **blocked**, i.e., not in spanning tree, will not forward data (prevent looping)
- root and designated ports are **active** – **forward** traffic

Spanning Tree example

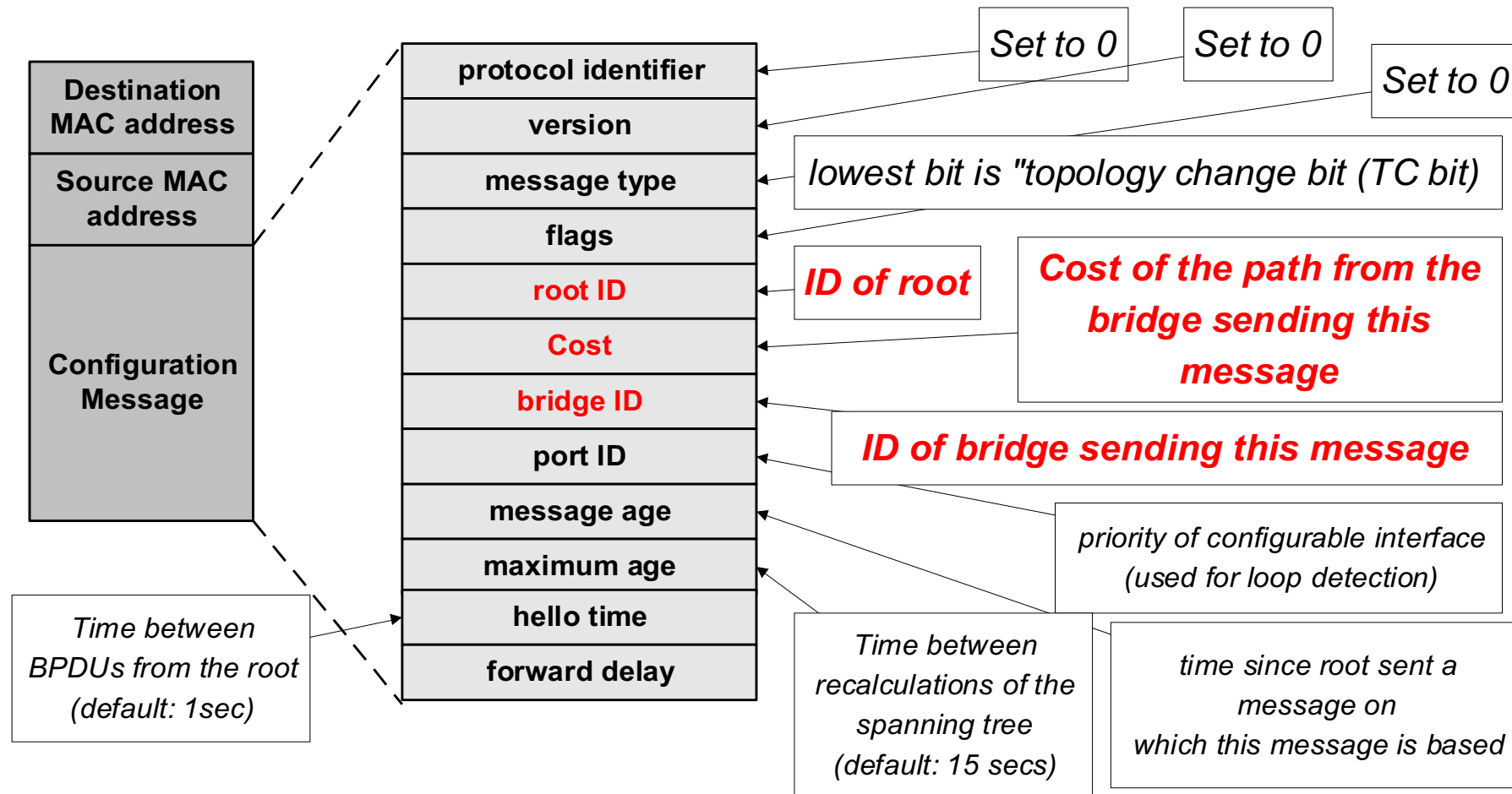


What do the BPDUs do?

with the help of the BPDUs, bridges can:

- elect a single bridge as the **root bridge**.
- calculate the distance of the **shortest path** to the root bridge
- each LAN can determine a **designated bridge**, which is the bridge closest to the root on a shared medium (e.g., Ethernet). The designated bridge will forward packets towards the root bridge.
- each bridge can determine a **root port**, the port that gives the best path to the root.
- select ports to be included in the spanning tree (**Active and Passive/Blocked**).

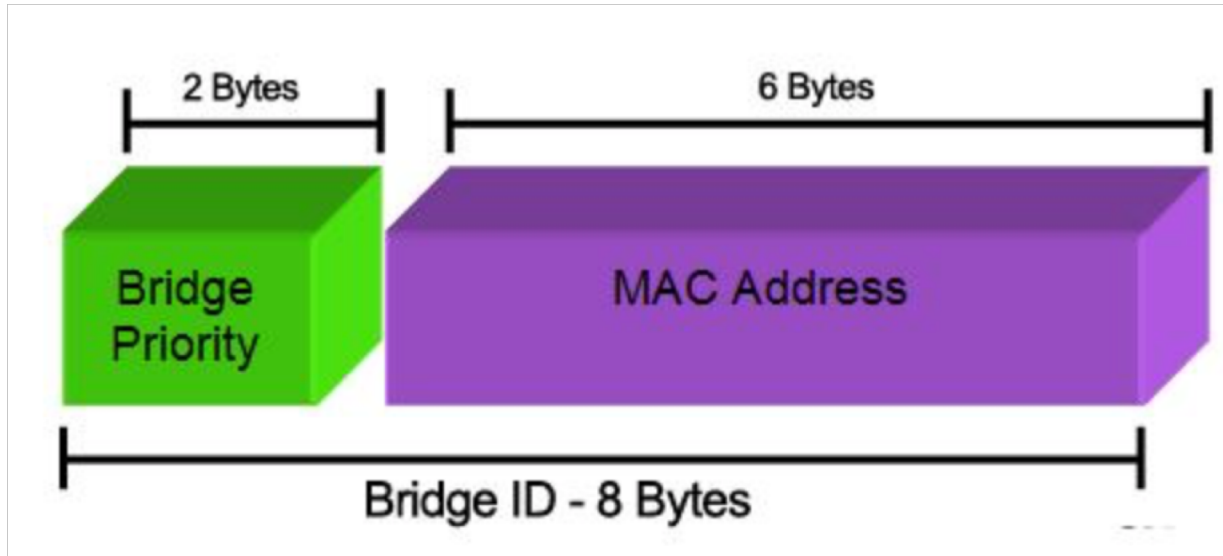
Configuration BPDUs



Concept

- Each bridge as a unique identifier:

Bridge ID = <priority level.MAC address>



- Note that a bridge has several MAC addresses (one for each port), but only **one ID**. It picks the **lowest MAC address**
- Each port within a bridge has a unique identifier - **port ID**.
- Root Bridge:** The bridge with the lowest ID is the root of the spanning tree. I.e., lowest MAC and priority level.
- Root Port:** Each bridge has a root port which identifies the next hop from that bridge to the root. This port is identified during the spanning tree process

Building a Spanning Tree

- **Root Path Cost:**

For each bridge, the cost of the min-cost path to the root.

Usually it is measured in #hops to the root

- **Designated Bridge, Designated Port:**

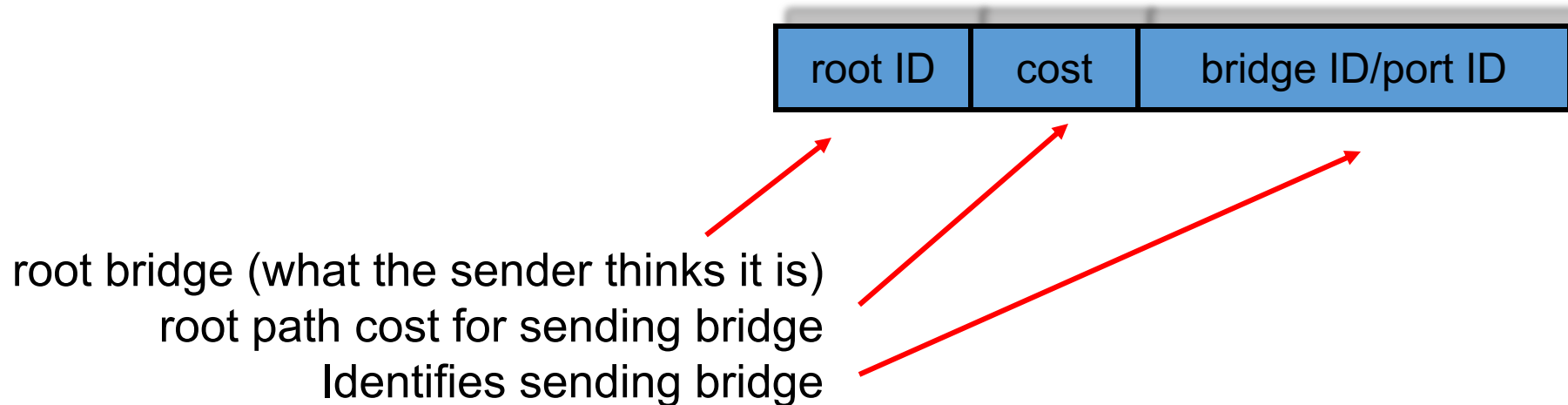
- Single bridge on a LAN that provides the minimal cost path to the root bridge for this LAN:

- if two bridges on a LAN have the same cost, select the one with lowest ID
- if the min-cost bridge has two or more ports on the LAN, select the port with the **lowest port ID**

Steps of Spanning Tree Algorithm

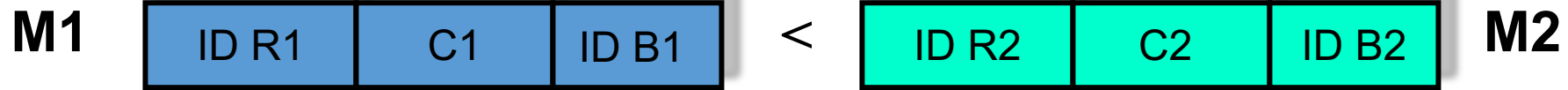
1. **Determine the root bridge**
2. **Determine the root port on all other bridges**
3. **Determine the designated port on each LAN**

Each bridge sends out BPDUs that contain the following information:



Ordering of Messages

- We can order BPDU messages with the following ordering relation “ $\ll$ ”:



If $(R1 < R2)$

$M1 \ll M2$

elseif $((R1 == R2) \text{ and } (C1 < C2))$

$M1 \ll M2$

elseif $((R1 == R2) \text{ and } (C1 == C2) \text{ and } (B1 < B2))$

$M1 \ll M2$

- If above holds, M1 is dominant and M2 will change its information to match M1
- Else M2 is dominant and M1 changes to follow M2

Determine the Root Bridge

- initially, all bridges assume they are the root bridge.
- each bridge B sends BPDUs of this form on its LANs:



- each bridge looks at the BPDUs received on all its ports and its own transmitted BPDUs.
- root bridge, at any point in time, is the **smallest received root ID** that has been received so far.
 - Whenever a smaller bridge ID arrives, the root field is updated in a bridge's BPDU root bridge field
 - Otherwise bridge maintains the current value

Calculate the Root Path Cost - Determine the Root Port

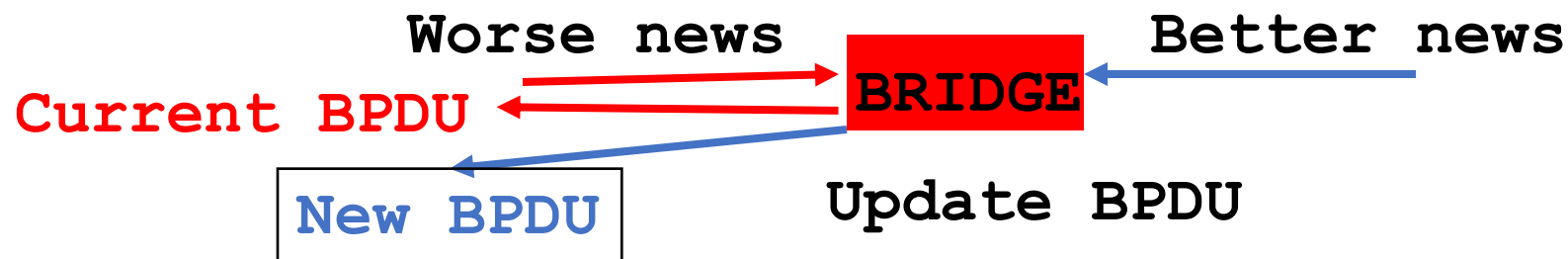
- at some point bridge B has a **belief** of who the root is, say R.
- bridge B determines the Root Path Cost (Cost) as follows:
 - **if $B = R$** : Cost = 0 (i.e., it is the root)
 - **if $B \neq R$** : Cost = {Lowest Cost found in a received BPDU with R as root} + 1

Updating and Sharing Information

- **B's root port** is the port from which **B** received the lowest cost path to R (in terms of relation " $\leq$ "). E.g., **port A**
- knowing **R** and **Cost**, **B** can generate a current BPDU with its root port **A**.

R	Cost	B	A
---	------	---	---

- bridge B will only **send** its current BPDU to a **neighbor** if it receives "**worse news**" on any of its ports from that neighbor.
- it **updates** its BPDU if it receives "**better news**" from a neighbor on one of its ports (that port now becomes the root port) and broadcasts an updated "**new**" BPDU on all its non root ports.

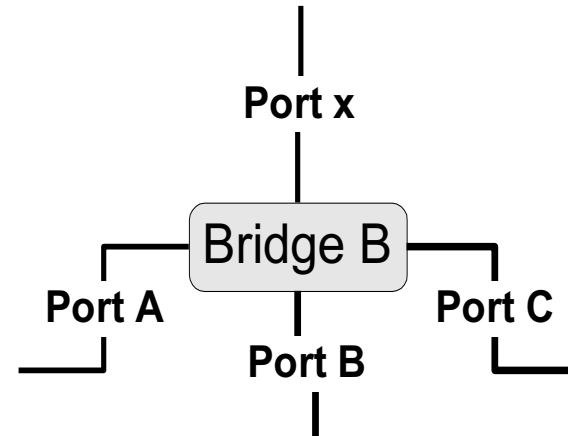


In Summary

- now that B has generated an updated BPDU, it shares the new information on all of its “**other**” ports



- B will send this BPDU on all of its ports, except port **X**, **if X is port from which it received better news.**
- in this case, B also assumes that it is the **designated bridge** for the LAN to which port **X** connects.
- and **Port X** is the **root port**.



Example

- bridge **B (ID 8)** receives on port “X” the following BPDU from a neighboring bridge (**ID 12**):

R=2	Cost=3	12	C
-----	--------	----	---

- and **B**’s current BPDU is:

R=2	Cost=2	8	A
-----	--------	---	---

- because Cost **2** << Cost **3**, **B** will broadcast on its port “**X**” its **current** BPDU to let bridge (**ID 12**) know it has a shorter path to the root bridge (**ID 2**) via port **A** (its current **root** port).

- if instead **B**’s current BPDU is:

2	5	8	A
---	----------	---	---

- then **B** will broadcast on all its **other ports** (A,B,C) the following BPDU:

2	<u>3+1=4</u>	8	<u>X</u>
---	--------------	---	-----------------

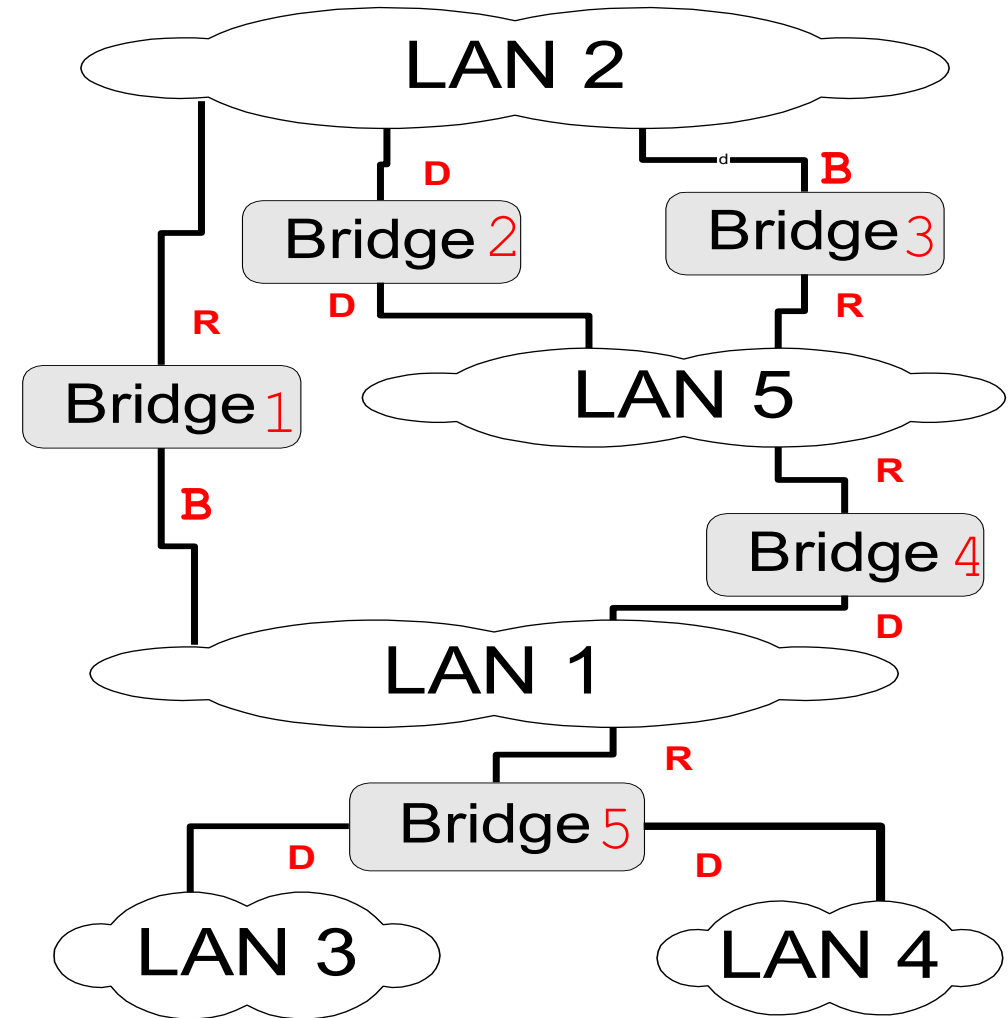
- where “**4**” (3+1) is new cost from Bridge B on port **X** to **root** bridge (**ID 2**) (via bridge **ID 12**). And port **X** is now its **root** port

Selecting the Ports for the Spanning Tree

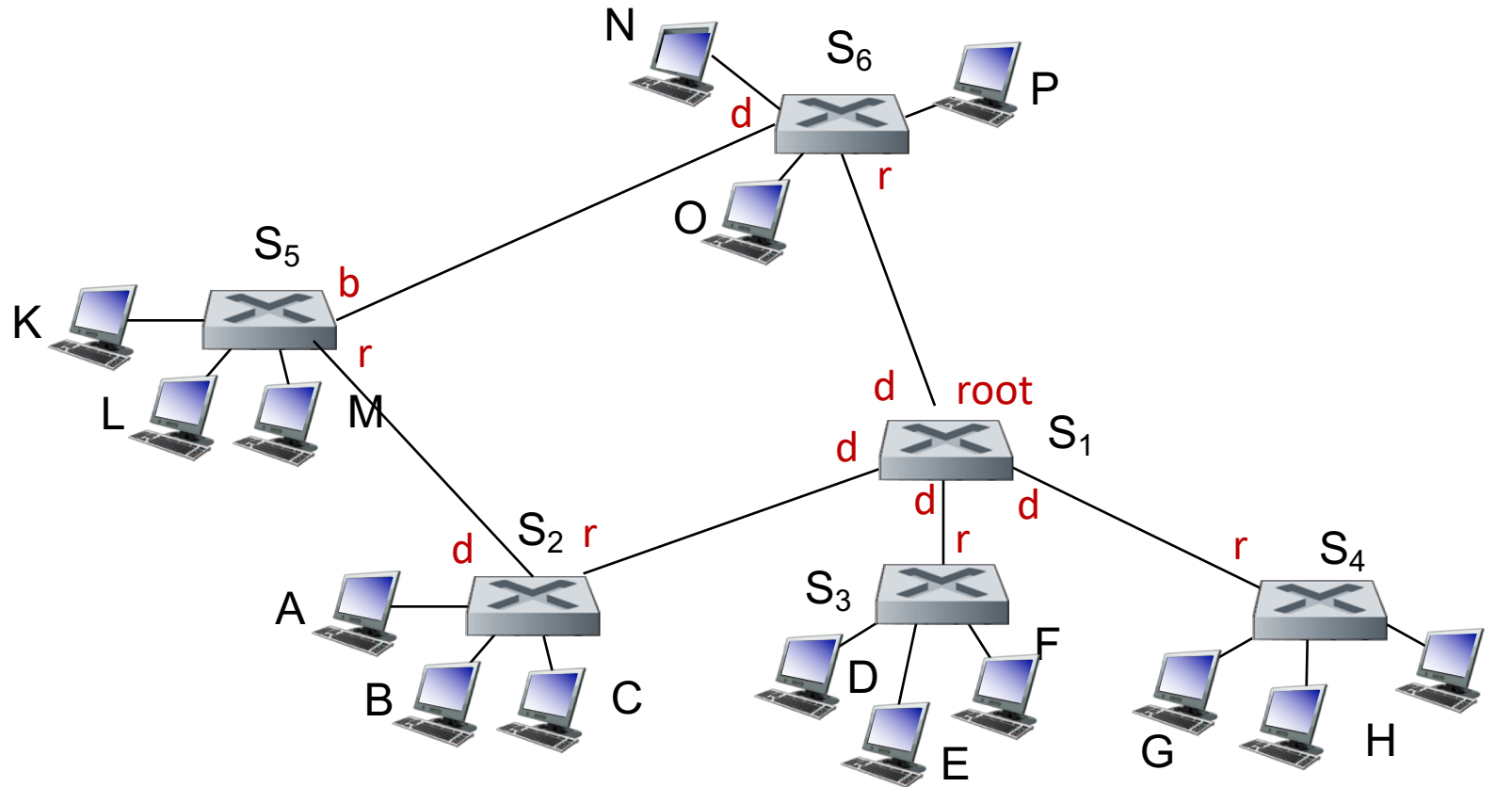
- now that Bridge B has calculated the root, the root path cost, and the designated bridge and port for each LAN.
- **B can decide which ports are in the spanning tree:**
 - B's **root port** is part of the spanning tree (every bridge has to have a root port)
 - The ports on **all** the LANs for which B is the **designated** bridge are part of the spanning tree (**designated ports**).
- B's ports that are in the spanning tree will forward packets (**=forwarding state**)
- B's ports that are not in the spanning tree will not forward packets (**=blocking state**)
- note that a bridge may **not be** the designated bridge on **any** LAN. As such, **all its ports**, other than the **root port** will be **blocked**.

Spanning Tree example – what is the spanning tree?

- consider the network on the right.
- assume that the bridges have calculated the designated ports (**D**), the blocked ports (**B**) and the root ports (**R**) as indicated.
- what is the spanning tree?

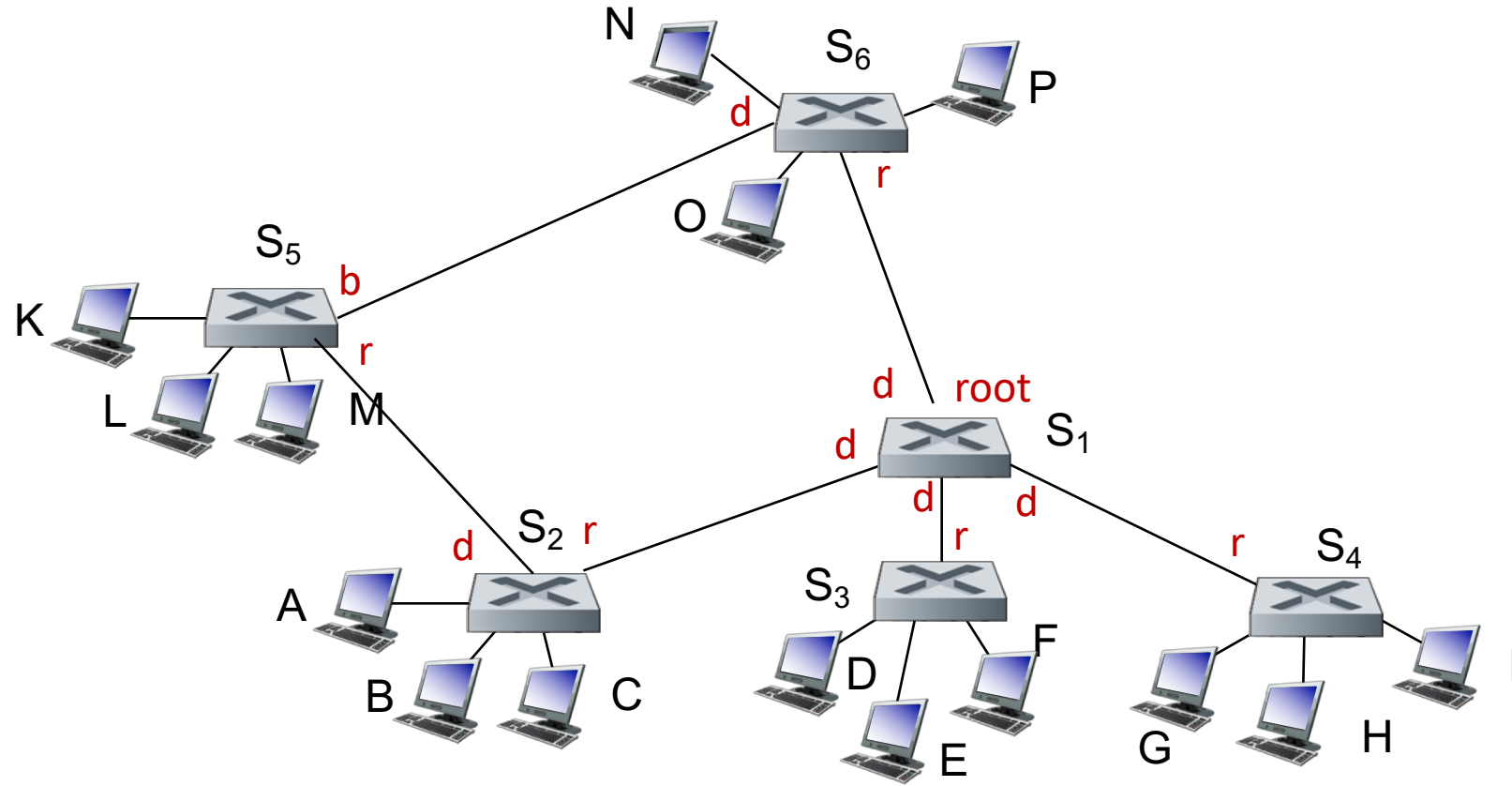


Spanning Tree example – what is the spanning tree?



Health of the Spanning Tree

- consider the network on the right - if link between S_5 and S_2 goes down or S_2 fails -> tree is damaged
- S_5 is now isolated and cannot send to, or receive traffic from, the rest of the tree
- using “hello” messages, the switches will notice the failure of a component and re-calculate the spanning tree



Detecting and Recovering from Failures

- *hello* messages are generated periodically by the root switch, which sends them out on each link to the next level of switches
- every switch that receives a *hello* message, replaces the bridge ID with its own, and passes it on to the next level of switches
- all switches receive the *hello* messages on their root ports, and forward them via their designated ports
- if a switch does not receive a hello message within a certain period of time (fixed interval), it starts a “recovery” timer that is 3x the *hello* message interval
- if that timer expires, it will assume the tree is broken and initiate the “spanning tree” process
- when a problem is detected by a switch (i.e., timer expired), it resets all its ports and initiates the spanning tree algorithm by sending out a BPDU declaring itself as Root.

